Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/27/2006
PI Name:	Milstead, Jeffrey		
Project Title:	Comparison of Space Flight and Hindlimb Suspension Skeletal Unloading Models in Mice: Exploring Mechanisms of Mineralization		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Physiology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	UNIVERSITY	**Phone:**	864-656-4466
Organization Name:	Clemson University		
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PI Web Page:			
City:	Clemson	**State:**	SC
Zip Code:	29634-0905	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	GSRP
Start Date:	07/01/2004	**End Date:**	06/30/2006
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA HQ
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Spaceflight creates an environment that causes a profound and accelerated osteoporosis in humans. Studies indicate that spaceflight experiments with mice hold promise in predicting spaceflight effects on humans. Analysis and comparison of skeletal unloading models in mice, including spaceflight and hindlimb suspension, will be performed in order to determine the accuracy of ground based osteopenia models. Engineering analysis will be coupled with biologic mechanisms to determine what is occurring to bone under unloaded conditions. Once the predictive nature of hindlimb suspension in mice has been characterized, research will focus on the inhibition of mineralization that occurs during skeletal unloading.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated:)